Math 257 Definitions and Theorems

Week 4: Modules 12-14 and Spring-Mass Systems

Fall 2023

12 LU Decomposition

Definition 12.1. An $n \times n$ matrix A is called

- **upper triangular** if it is of the form

$$\begin{bmatrix} \star & \star & \star & \star & \star \\ 0 & \star & \star & \star & \star \\ 0 & 0 & \star & \star & \star \\ 0 & 0 & 0 & \ddots & \vdots \\ 0 & 0 & 0 & 0 & * \end{bmatrix},$$

- **lower triangular** if it is of the form

$$\begin{bmatrix} \star & 0 & 0 & 0 & 0 \\ \star & \star & 0 & 0 & 0 \\ \star & \star & \star & 0 & 0 \\ \star & \star & \star & \ddots & \vdots \\ \star & \star & \star & \star & \star \end{bmatrix}.$$

Theorem 12.1. *The product of two lower (upper) triangular matrices is lower (upper) triangular.*

Remark. The inverse of a lower (upper) triangular matrix (if it exists) is again lower (upper) triangular.

Definition 12.2. A matrix A has an **LU decomposition** if there is a lower triangular matrix L and a upper triangular matrix U such that $A = LU$.

Theorem 12.2. *Let A be an $n \times n$-matrix. If A can be brought to echelon form just using row operations of the form $R_i \to R_i + cR_j$ where $j < i$, then A has an LU-decomposition.*

Remark. Not every matrix has an LU decomposition. The LU decomposition of a matrix is not unique.

13 Solving Linear Systems using LU Decomposition

Theorem 13.1. *Let A be an $n \times n$-matrix such that $A = LU$, where L is a lower triangular matrix and U is a upper triangular matrix, and let $\mathbf{b} \in \mathbb{R}^n$. In order to find a solution of the linear system*

$$A\mathbf{x} = \mathbf{b},$$

it is enough to find a solution of the linear system

$$U\mathbf{x} = \mathbf{c},$$

where $\mathbf{c}$ satisfies $L\mathbf{c} = \mathbf{b}$.

Theorem 13.2. *Let A be $n \times n$ matrix. Then there is a permutation matrix P such that PA has an LU-decomposition.*

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See the notes for information about this application.

14 Inner Product and Orthogonality

Definition 14.1. The **inner product** of $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is

$$\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T\mathbf{w}.$$

Theorem 14.1. *Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^n$, and let c be any scalar. Then*

(a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

(b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

(c) $(c\mathbf{u}) \cdot \mathbf{v} = c\,(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$

(d) $\mathbf{u} \cdot \mathbf{u} \geq \mathbf{0}$, *and* $\mathbf{u} \cdot \mathbf{u} = \mathbf{0}$ *if and only if* $\mathbf{u} = \mathbf{0}$.

Definition 14.2. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$.

- The **norm** (or **length**) of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}.$$

- The **distance** between $\mathbf{v}$ and $\mathbf{w}$ is

$$\operatorname{dist}(\mathbf{v}, \mathbf{w}) = \|\mathbf{v} - \mathbf{w}\|.$$

Definition 14.3. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. We say $\mathbf{v}$ and $\mathbf{w}$ are **orthogonal** if $\mathbf{v} \cdot \mathbf{w} = 0$.

Theorem 14.2. *Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be non-zero. Then $\mathbf{v}$ and $\mathbf{w}$ are orthogonal if and only if they are perpendicular (ie. if they form a right angle).*

Definition 14.4. A set of vectors in $\mathbb{R}^n$ is **pairwise orthogonal** if each pairing of them is orthogonal. We call such a set an **orthogonal set**.

Definition 14.5. A **unit vector** in $\mathbb{R}^n$ is vector of length 1.

Definition 14.6. A set of vectors form an **orthonormal set** if this set is an orthogonal set and all vectors in the set are unit vectors.